X-17
(2030.17)
L.N. STABILIZER BTMS./L.N. FEED
EXCHANGER
12" I.D. x 192" L
0.78 MM BTU/HR
DESIGN- SHELL: 160 PSIG @ 400° F
TUBES: 225 PSIG @ 300° F

X-18
(2030.18)
LT. NAPHTHA
COOLER
13" I.D. x 192" L
0.75 MM BTU/HR
DESIGN- SHELL: 160 PSIG @ 300° F
TUBES: 150 PSIG @ 250° F

X-16
(2030.16)
LT. NAPHTHA STABILIZER
REBOILER
18" I.D. x 156" L
1.42 MM BTU/HR
DESIGN- SHELL: 200 PSIG @ 600°F
TUBES: 275 PSIG @ 700°F

V-16
(2010.16)
LT. NAPHTHA STABILIZER
TOWER
3'-0" I.D. x 64'-7" T/T
-DESIGN-
150 PSIG @ 650° F

V-30
(2010.30)
LT. NAPHTHA STABILIZER
OVERHEAD ACCUMULATOR
4'-0" I.D. x 10'-0" T/T
-DESIGN-
300 PSIG @ 130° F

X-39
(2030.39)
LT. NAPHTHA STABILIZER
OVERHEAD CONDENSER
14" I.D. x 192" L
1.0 MM BTU/HR
DESIGN- SHELL: 300 PSIG @ 250° F
TUBES: 125 PSIG @ 175° F

(2017.04) PSV 04
SIZE: 3"xKx4"
BASIS:
SET: 150 PSI

TO FLARE 6402-D 2001 SHT. 1 240
TO FUEL GAS 6402-D 2002 SHT. 3 416
SOUR GAS

COOLING WATER RETURN 6602-D-2000 SHT. 6 621
TO COOLING TOWER #1 (NORTH TOWER)

FROM V-15 11 2002-D 2001 SHT. 8
TO V-300 6602-D 200 SHT. 2 12
TO FV-133 2002-D 2001 SHT. 11 63
STAB. LT. NAPHTHA

COOLING WATER SUPPLY 6002-D-2000 SHT. 7 622
FROM COOLING TOWER #1 (NORTH TOWER)

FM. P-37, 60 2002-D 2001 SHT. 4 28
TOP CRUDE

FM. P-403A/B 3102-D 2000 SHT. 9 220

FM. P-57,61,65 2002-D 2001 SHT. 8 46
LT. NAPHTHA

TO X-43 2002-D 2001 SHT. 12 47
TOP CRUDE

TO X-47 2002-D 2001 SHT. 11 80
STABILIZED LT. NAPHTHA

FROM X-47 81 2002-D 2001 SHT. 11
STABILIZED LT. NAPHTHA

FM. 150# STEAM OSBL

2" COOLING WATER RETURN
2" COOLING WATER SUPPLY
2"-BY-PASS
2"-STEAM OUT
SAMPLE
DRAIN
WEIR
IAS

P-58
(2026.58)
L.N. STABILIZER REFLUX
PUMP "SPARE"
DESIGN: 30 GPM
Δ HEAD: 245'
SP. GR.: 0.60

P-64
(2026.64)
L.N. STABILIZER REFLUX
PUMP
DESIGN: 30 GPM
Δ HEAD: 245'
SP. GR.: 0.60

REV. NO.	DATE	REVISED	REV. BY	CKD. BY	APP'D
35	10/11/19	MOC 19-32	HRP	CS	–
34	08/30/19	MOC 19-33	HRP	BJH	–
33	03/28/18	MOC 18-74	RS	CF	DD
32	06/21/17	MOC 17-18	DJA	DRS	–
31	08/15/16	MOC 16-19	LMM	DRS	–
39	12/31/24	BORDER SWAP	HRP	–	–
38	02/04/22	MOC 20-18 AS-BUILT	–	TS	–
37	01/25/22	MOC 21-75 AS-BUILT	HRP	BJH	–
36	07/15/20	MOC 20-19	LMM	DRS	DLP

KERN ENERGY
We Fuel the Future

DRAWN BY: S.CALKINS	DATE: 05/30/07	
DESIGN:	DATE:	
CHECK:	DATE:	
ENGINEER:	DATE:	
APPROVED:	DATE:	
APPROVED:	DATE:	
APPROVED:	DATE:	

PIPING & INSTRUMENTATION DIAGRAM
CRUDE UNIT
KERN ENERGY
BAKERSFIELD,CA. 93307

SCALE:	DWG NO:	SH:	REV. NO.
N.T.S	2002-D 2001	9 of 16	39

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